CHRIST (Deemed to be University), Bangalore – 560 029

Department of Mathematics

END SEMESTER EXAMINATION – April/May 2024

UG IV Semester

Programme Name: BCA **Max. Marks: 100**

Course Name: Graph Theory **Time: 3 Hrs**

Course Code: BCA431

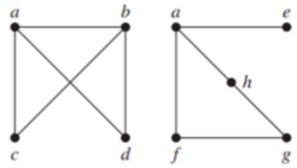
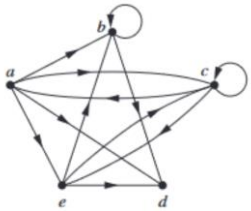
General Instructions

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / Course title & number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place in the hall.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer booklet. No sheets should be detached from the answer booklet.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer booklet.

Course Outcomes (COs): The students will able to
CO1: Enhance research, inquiry and analytical thinking abilities.
CO2: To improve proof-writing skills.
CO3: Apply the basics of Graph Theory to solving practical problems.

SECTION A

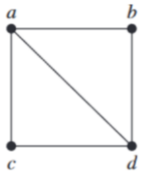
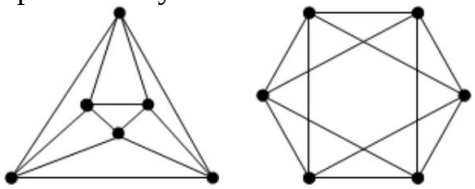
Answer ALL the questions **10 × 3 = 30 Marks**

Q. No	Questions	CO	RBT
1	Describe a cubic graph. What is the number of vertices in a cubic graph with 15 edges?	CO2	L3
2	What is meant by Collaboration Graph? Give example.	CO2	L2
3	Find the union of the given pair of simple graphs. 	CO2	L2
4	Use adjacency list to describe the directed graph given below. 	CO2	L3
5	Distinguish Eulerian and Hamiltonian graphs. Give an example of a graph that is neither Eulerian nor Hamiltonian.	CO1	L4

6	Define edge chromatic number of a graph. Find the edge chromatic number of K_4. **OR** State Euler's formula for planar graphs and verify the same for the following graphs. (i) K_4 (ii) W_4.	CO3	L5
7	Discuss (i) Radius of the graph. (ii) Diameter of the graph. Evaluate the radius and diameter of K_6 and C_6.	CO3	L5
8	Define (i) tree (ii) rooted tree (iii) subtree with examples. **OR** How many leaves does a full 3 – ary tree with 100 vertices have?	CO2	L4
9	Evaluate the value of the prefix form $* +3+3 \uparrow 3+3\ 3\ 3$.	CO3	L4
10	Define a spanning tree. Also, find a spanning tree of (i) $K_{2,3}$ (ii) W_5.	CO2	L3

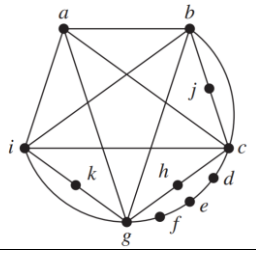
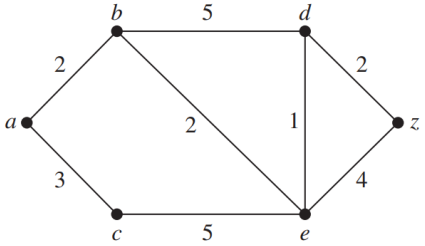
SECTION B

Answer ALL the questions **3 × 7 = 21 Marks**

Q. No	Questions	CO	RBT
11	Define (i) cut vertices (ii) cut edges (iii) cut set with appropriate examples.	CO2	L4
12	i) Prove that an undirected graph has even number of vertices of odd degree. ii) Draw the graph represented by the given adjacency matrix A. $A = \begin{bmatrix} 0 & 1 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 2 & 1 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$	CO2	L4
13	Find the number of paths of length four from the vertex a to vertex d in a simple graph G given by  **OR** Explain the isomorphism of graphs. Examine whether the following graphs are isomorphic. Justify. 	CO3	L5

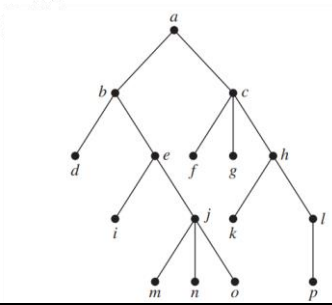
SECTION C

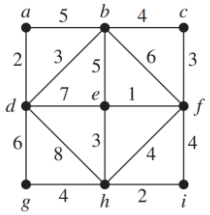
Answer ALL the questions **4 × 7 = 28**

Q. No	Questions	CO	RBT
14	Explain Konigsberg's bridge problem and Euler's contribution in solving it in detail.	CO1	L2
15	Describe homeomorphic between two graphs. Determine whether the given graph is plannar. 	CO3	L3
16	Discuss the algorithm for coloring a graph based on vertex degrees and adjacency relationship with example.	CO2	L2
17	Find the length of a shortest path between a and z in the given weighted graph by using Dijkstra's algorithm  **OR** Discuss the applications of graph colorings on (i) Scheduling Final Exams and (ii) Index Registers.	CO3	L4

SECTION D

Answer ALL the questions **3 × 7 = 21**

Q. No	Questions	CO	RBT
18	Define universal address system. For a given ordered rooted tree T, find the preorder traversal and postorder traversal. 	CO1	L3

19	Explain Prims algorithm and use it to obtain minimal spanning tree of the weighted graph given below:  	CO2	L2
20	Use Huffman coding to encode the following symbols with frequencies listed below. $A: 0.10$, $B: 0.25$, $C: 0.05$, $D: 0.15$, $E: 0.30$, $F: 0.07$, $G: 0.08$. What is the average number of bits used to encode a character? **OR** Explain N−queens problem and use backtracking to solve $N-$ queens problem for $N = 4$.	CO3	L4

Revised Bloom's Taxonomy (RBT) Levels:		
L1 – Remembering	L2 – Understanding	L3 – Applying
L4 – Analyzing	L5 – Evaluating	L6 - Creating